

Problem 1. SVM.

In this problem, you will prove that the hard-SVM rule, namely,

$$\operatorname{argmax}_{(w,b): \|w\|=1} \min_{i \in [m]} |\langle w, x_i \rangle + b| \quad \text{s.t.} \quad \forall i, \ y_i(\langle w, x_i \rangle + b) > 0,$$

is equivalent to the following formulation:

$$\operatorname{argmax}_{(w,b): \|w\|=1} \min_{i \in [m]} y_i(\langle w, x_i \rangle + b).$$

1. Define $\mathcal{G} := \{(w, b) : \forall i, \ y_i(\langle w, x_i \rangle + b) > 0\}$. Show that

$$\operatorname{argmax}_{(w,b): \|w\|=1} \min_{i \in [m]} y_i(\langle w, x_i \rangle + b) \in \mathcal{G}.$$

2. Show that for any $(w, b) \in \mathcal{G}$,

$$\min_{i \in [m]} y_i(\langle w, x_i \rangle + b) = \min_{i \in [m]} |\langle w, x_i \rangle + b|,$$

and finish the proof of the equivalence.

Solution.

Problem 2. Rademacher Complexity.

In this problem, you will prove the Massart lemma:

Lemma 1 (Massart lemma). *Let $A = \{a_1, \dots, a_N\}$ be a finite set of vectors in $\mathbb{R}^m$. Define $\bar{a} = \frac{1}{N} \sum_i a_i$. Then, the Rademacher complexity of A can be bounded by:*

$$R(A) \leq \max_{i \in [N]} \|a_i - \bar{a}\| \cdot \frac{\sqrt{2 \log N}}{m}.$$